C-H-111d
February 15, 1968
SUPERSEDING
Int. Fed. Spec. C-H-00111c(GSA-FSS)
August 28, 1967 and
Fed. Spec. C-H-111a
September 14, 1948

FEDERAL SPECIFICATION

HAIR, ANIMAL, CURLED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for curled hair, primarily intended for use in the manufacture of furniture and mattresses.

1.2 Classification.

1.2.1 Types. The curled hair covered by this specification shall be of one grade and of the following types as specified (see 6.1).

Type I - Horse-tail hair.
Type II - Cattle-tail hair.
Type III - 75 percent Cattle-tail hair and 25 percent winter hog hair.
Type IV - 50 percent Cattle-tail hair and 50 percent winter hog hair.
Type V - 30 percent Cattle-tail hair and 70 percent winter hog hair.
Type VI - Horse-mane hair.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-P-268 - Paper, Kraft, Untreated, Wrapping.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-T-60 - Tape; Pressure-Sensitive Adhesive, Waterproof, for Packaging.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

FSC 8320

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The hair used in the manufacture of curled hair conforming to this specification shall be resilient new stock. The material shall not contain vegetable fibers, fetlock hair, hair-cloth clippings, combings, screenings, used or reworked hair, or hair from animals other than specified by the type descriptions. Horse-mane and horse-tail hair shall be clipped. Combings may be used for **types I and VI animal hair.**

3.2 Cleaning. The hair shall be thoroughly washed, all epidermis, scaly matter, dust and foreign matter removed and be permanently deodorized before spinning. The finished hair shall have no objectional odor and shall impart no odor or visible contamination to the water extract when tested as specified in 4.3.4 and 4.3.4.1.

3.3 Curl setting and sterilization. The hair in the form of ropes shall be steamed with saturated steam at 20 pounds pressure (approximately 259°F.) for not less than 20 minutes, or actively boiled in water for not less than 1 hour, in a manner to assure uniform temperature throughout the material. It shall be subsequently stored for not less than 15 days in a manner to assure uniform drying and then carded (or picker picked).

3.3.1 In the curl setting process, conditions shall be such that prime winter hog hair shall be curled to the shape of the letters "c", "j", "u" or "o" or a corkscrew twist having at least a half-circle form within the spiral loop.

3.4 Color. The color shall be natural. The curled hair shall show no presence of dye when tested as specified in 4.3.4.2.

3.5 Length.

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3.5.1 Type I. The length of the curled and carded type I horse-tail hair shall be that which is used in the best commercial practice for the purpose specified in 1.1.

3.5.2 Type II. The length of the curled and carded type II cattle-tail hair shall be in accordance with the following table:

Not more than 12 percent by weight under 2 inches in length.
Not more than 50 percent by weight under 6 inches in length.
Not more than 90 percent by weight under 10 inches in length.
Not less than 10 percent by weight 10 inches and over in length.

3.5.3 Type III. The length of the curled and carded type III mixture of cattle-tail and winter hog hair shall be in accordance with the following table:

Not more than 32-1/2 percent by weight under 2 inches in length.
Not more than 62 percent by weight under 6 inches in length.
Not more than 93 percent by weight under 10 inches in length.
Not less than 7 percent by weight 10 inches and over in length.

3.5.4 Type IV. The length of the curled and carded type IV mixture of cattle-tail and winter hog hair shall be in accordance with the following table:

Not more than 55 percent by weight under 2 inches in length.
Not more than 74 percent by weight under 6 inches in length.
Not less than 7 percent by weight 9 inches and over in length.

3.5.5 Type V. The length of the curled and carded type V mixture of cattle-tail and winter hog hair shall be in accordance with the following table:

Not more than 73 percent by weight under 2 inches in length.
Not more than 90 percent by weight under 5 inches in length.
Not less than 10 percent by weight 5 inches and over in length.

3.5.6 Type VI. The length of the curled and carded type VI horse-mane hair shall be that which is used in the best commercial practice for the purpose specified in 1.1.

3.6 Weight. Weight for purposes of acceptance shall be net dry weight as determined by correction for dirt and moisture as specified in 4.4, except that when dirt and moisture content exceed 10 percent, the excess over 10 percent shall be doubled in computing the deduction.

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3.7 Workmanship. The finished hair shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed that established by the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 In-process inspection. In-process inspection shall be made to assure that processing is conducted in accordance with specified requirements. The supplier shall certify that material and processing requirements have been met.

4.3 Inspection of the end item. Inspection shall be in accordance with MIL-STD-105 except where otherwise indicated.

4.3.1 Sampling. Unless otherwise specified, at the place of manufacture, a composite sample shall be made up with approximately 1 pound samplings from each 500 pound portion of the lot or fraction thereof offered for delivery. If the lot exceeds 5,000 pounds, ten 1 pound samplings shall be taken from equal portions of the lot. The samplings shall be combined and thoroughly mixed to assure that the combined sample represents the composite of the lot.

4.3.2 Visual examination. The composite sample shall be examined and defects classified as listed below. Any defect found shall be cause for rejection.

Presence of hair other than specified for the required type (see 4.3.2.1).
Presence of vegetable fiber.
Hair not free of epidermis, scaly matter, dust and foreign matter.
Objectionable odor.
Hair not well curled and carded (or picker picked).
Hog hair (for applicable types) not curled as specified (see 3.3.1).

4.3.2.1 Horse-tail hair. In case of dispute, horse-tail hair may be distinguished from horse-mane hair by measuring the average diameter with the aid of a microscope. Horse-tail hair shall measure not less than 0.0055 inch.

4.3.3 Examination for length. The composite sample shall be carefully opened and two specimens of 5.00 grams each shall be taken at random for test. Hairs shall be picked out individually and measured and sorted to applicable length groupings for the required type (see 3.5). Each length grouping shall then be weighed, and its percentage of the total specimen shall be calculated. The average of the results of each length group in the two specimens shall represent the lot. The lot shall be unacceptable if any length group is not within the applicable limits established for the required type.

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4.3.4 Tests. Testing of the curled hair shall be performed in accordance with 4.3.4.1 or 4.3.4.2. The sample for testing purposes shall be the composite sample made up as specified in 4.3.1. Any failure is cause for rejection of the lot.

4.3.4.1 Odor and visible contamination. Approximately 4 ounces of hair taken from the composite sample shall be immersed in 2 quarts of sterile distilled water for a period of not less than 2 hours. The mixture shall then be thoroughly agitated by stirring and the hair shall then be strained out with a 14-mesh screen. The water extract shall be immediately examined for clarity and freedom from sediment. The water extract shall then be allowed to stand in an unsealed container for approximately 72 hours at room temperature (70° to 80°F.) and then checked for odor. Visible contamination or odor following standing shall be considered failure and shall be cause for rejection of the lot.

4.3.4.2 Artificial coloration. The presence of dye in the hair shall be determined by gently boiling a 5-gram specimen of the hair in 250 cubic centimeters of a 20 percent (by volume) solution of hydrochloric acid in a 400-cubic-centimeter beaker. If dyed hair is present a pink coloration will appear in a few minutes, the density of the coloration depending upon the amount of dyed hair present. A 25-gram specimen of hair shall also be boiled in a 2 percent solution of sodium carbonate or a 3 percent solution of ammonium hydroxide for one-half hour. If dyed hair is present, the solution will be darker than if only natural hair were present. When the presence of dyed hair is detected by these methods, some of the hairs shall be cross sectioned by means of a freezing microtome and examined microscopically to determine conclusively that the hair is dyed. Evidence of artificial coloration by test or examination shall be cause for rejection of the lot.

4.4 Determination of net dry weight. At the time of weighing shipments at the place of delivery, weight samplings shall be taken from the interior of the bales according to the sampling plan described in 4.3.1. The combined samplings shall then be weighed immediately to the nearest ounce or one tenth pound. The combined sample shall then be washed with agitation in a 1 percent neutral soap solution at a temperature of 100°F. ± 5°F., and then rinsed with agitation in water at the same temperature until the rinse water is clear. The material shall then be drained of water and dried to constant weight at 212° to 220°F. and weighed. The percentage loss in weight from the sample shall be calculated to the nearest 0.5 percent and this percentage deducted from the "as weighed" weight of the lot in determining the net dry weight for acceptance (see 3.6 for additional deduction when over 10 percent).

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table 1. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II with an acceptable quality level (AQL) of 4 defects per hundred units.

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TABLE I. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A. The product shall be packed in compressed bales. Each bale shall be tied or banded with metal ties of wire or steel banding. Each compressed bale shall be wrapped in paper conforming to UU-P-271, class C-1. The paper wrap shall be sealed with tape conforming to PPP-T-60, class 1. The gross weight for each bale shall not exceed 100 pounds.

5.1.2 Level B. Fifty pounds of hair shall be packed in a 7-1/2 ounce burlap bag. The bags shall be sewn on the top and ends. When specified, the bag shall be lined with 60 pound minimum basis weight kraft paper conforming to UU-P-268, grade B.

5.1.3 Level C. The hair shall be packed to insure carrier acceptance and safe delivery at destination at the lowest rates, in containers complying with the rules and regulations applicable to the mode of transportation.

5.2 Marking

5.2.1 Civil agencies. In addition to markings required by the contract or order, the bales shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the bales shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of hair required (see 1.2.1).
(c) Selection of applicable packing levels (see 5.1).

6.2 Payment should be made on the basis of net weight delivered, unless the correction for dirt and moisture, determined as described in 4.4, exceeds 10 percent of the delivered weight, in which case twice the excess percentage should be deducted from the weight of the shipment for the purpose of payment.

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6.3 In order that the contractor may meet the requirements of 3.3 as to storing and carding and that there may be sufficient time for inspection, contracts for horsehair should not call for delivery before 30 days from date of contract or order, and purchasing officers should endeavor to anticipate their needs as far in advance as practicable and submit requests and requisitions accordingly.

MILITARY CUSTODIANS:

Army - GL
Air Force - 67
Navy -

CIVIL AGENCIES INTEREST:

GSA - FSS

Review activities:

Army - GL, MD
Air Force - 67
Navy -

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